CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Theory Examination (Supplementary Exam - 1) January 2024
7th Semester of B.Tech. (CS)

Design of Language Processor [CS450]

Marks: 70 **Duration: 195 mins.**

Section - 1

Answer all the questions. Section Duration: 40 mins

1. Which technique is used for implementing lexical analyzers efficiently? (2)
   - Recursive Descent Parsing
   - LL(1) Parsing
   - Regular Expressions
   - LR Parsing

2. Which of the following is not a part of the front-end of a compiler? (2)
   - Lexical Analyzer
   - Syntax Analyzer
   - Code Generator
   - Semantic Analyzer

3. What is the primary role of the Lexical Analyzer in a compiler? (2)
   - Generating intermediate code
   - Converting source code to tokens
   - Checking for semantic errors
   - Optimizing the generated code

4. In compiler design, what is the purpose of the Dead Code Elimination optimization? (2)
   - Removing unreachable code
   - Optimizing loop structures
   - Eliminating redundant instructions
   - Minimizing register usage

5. What is the primary function of the Code Generation phase in a compiler? (2)
   - Generating machine-independent code
   - Optimizing the intermediate code
   - Transforming the source code into tokens
   - Producing the final machine code

6. Which phase of the compiler is responsible for recognizing the basic syntactic structure of the source program and transforming it into a sequence of tokens? (2)
   - Lexical Analysis
   - Syntax Analysis
   - Semantic Analysis
   - Code Generation

7. Which of the following is not a compiler optimization technique? (2)
   - Loop Unrolling
   - Dead Code Elimination
   - Variable Declaration
   - Constant Folding

8. What is the purpose of the Semantic Analysis phase in a compiler? (2)
   - Generating machine code
   - Checking for syntax errors
   - Assigning types to expressions
   - Producing intermediate code

9. Which data structure is commonly used to implement a symbol table in a compiler? (2)
   - Stack
   - Queue
   - Hash Table
   - Binary Search Tree

10. Which phase of the compiler is responsible for converting the high-level programming language into an intermediate code? (2)
    - Lexical Analysis
    - Syntax Analysis
    - Semantic Analysis
    - Intermediate Code Generation

Section - 2

Answer 5 out of 7 questions.

1. Construct syntax tree and DAG for following expression: (4)
   X = a * (b+c)- (b+c)* d

2			Explain error recovery strategies used by parser.	(4)
3			Write difference(s) between stack and heap memory allocation.	(4)
4			Explain various parameter passing methods.	(4)

5			Define token, lexeme and pattern. Identify the lexemes that makes up the tokens for the following code
			cost p = 10;
			if(a < p)
			{
			a++ ;
			If(a== 5)
			continue ;
			}
				(4)

6			Give the rule to remove left recursive grammar. And Eliminate left recursion from following grammar. $S \rightarrow Aa \mid b$ $A \rightarrow Ac \mid Sd \mid f$	(4)
7			Write a short note on Symbol table management.	(4)

Section - 3

Answer 6 out of 7 questions.

1			Consider the following grammar: S' = S# S -> ABC A -> a|bbD B -> a| Ɛ C -> b| Ɛ D -> c| Ɛ Construct FIRST and FOLLOW for the grammar also design LL(1) parsing table for the grammar	(5)
2			Consider the following grammar S -> AA A -> aA A -> b And construct the LALR parsing table	(5)
3			What is DAG? What are its advantages in context of optimization? How does it help in eliminating common sub expression?	(5)
4			What is peephole optimization? Explain with example	(5)
5			Construct CLR parsing table for following grammar. S -> aSA | € A -> bS | c	(5)
6			Explain a rule of Left factoring a grammar and give Example.	(5)
7			Translate the following expression into quadruple, triple, and indirect triple: (a+b)*(c+d)-(a+b+c)	(5)

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